MILITARY STANDARD

MIL-STD-1320-7A (NAVY)

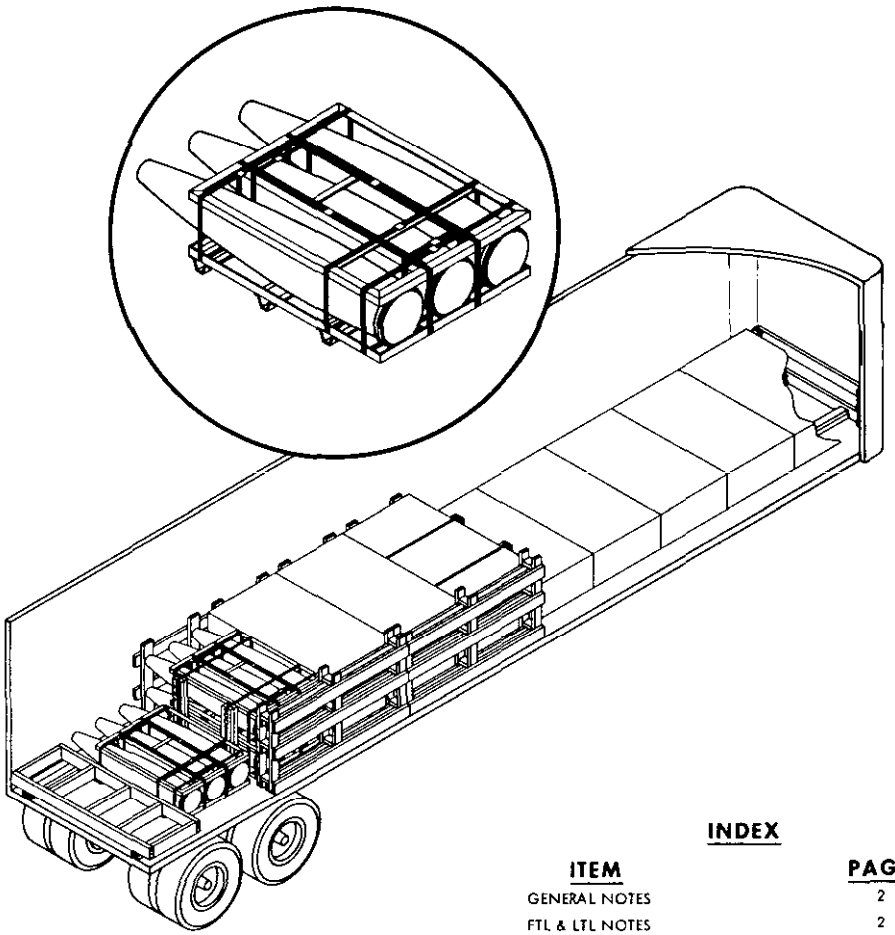
30 SEPTEMBER 1976

SUPERSEDING
MIL-STD-1320-7 (NAVY)
1 MAY 1976

TRUCKLOADING
BOMB, GENERAL PURPOSE MK 83 AND MODS (1000 LB)
IN PALLET ADAPTER MK 11 MOD 0
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

SEE TABLE I
PAGE 3



INDEX

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NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

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TECHNICAL DIRECTION AGENT (TDA) DATE

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE 1000 LB MK 83 AND MODS GENERAL PURPOSE BOMB IN BOMB PALLET MK 11 MOD 0. THE BOMBS MAY BE UNCOATED AND PALLETIZED IN ACCORDANCE WITH WR-54/35 OR THERMALLY PROTECTED AND PALLETIZED IN ACCORDANCE WITH WR-54/243. NO OTHER LOADS ARE AUTHORIZED FOR SHIPMENT PER THIS DOCUMENT.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44, AND 45 FT VAN TRAILERS WITH NAILABLE AND NON-NAILABLE FLOORS. THE TRAILER TANDEM AXLES SHALL BE LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. TO PREVENT LONGITUDINAL MOVEMENT OF THE UNIT LOADS IN THE SECOND LAYER, THE LOADS SHALL BE UNITIZED (SEE UNITIZING DETAIL, PAGE 6) WHERE THE PATTERN CHANGES FROM TWO LAYERS HIGH TO ONE LAYER HIGH. THE STIFFENERS SHALL FACE THE SINGLE LAYER SIDE.

NOTE

THE UNITIZED LOADS WILL WEIGH ALMOST 6000 LBS. IF ADEQUATE HANDLING EQUIPMENT IS NOT AVAILABLE, THE UNIT LOADS WILL HAVE TO BE PLACED INTO THE TRAILER ONE AT A TIME AND THE STACKED UNIT LOADS STRAPPED IN PLACE.

4. SIDE FRAME ASSEMBLY, DETAIL C, SHALL BE POSITIONED IN TRAILER WITH PIECE 12 AGAINST TRAILER WALL. TOENAIL TO FLOOR WITH TWO 16d NAILS PER JOINT TO PREVENT LONGITUDINAL MOVEMENT.

NOTE

WHEN TRAILERS HAVE NON-NAILABLE FLOORS, BLOCK OR TIE SIDE FRAME ASSEMBLIES TO UNIT LOADS TO PREVENT LONGITUDINAL MOVEMENT.

5. UNITIZED LOADS SHALL BE POSITIONED IN TRAILER WITH BOMB NOSES AGAINST PIECE 11 OF SIDE FRAME ASSEMBLY, DETAIL C.
6. WIDTH OF SIDE FRAME ASSEMBLY, DETAIL D, SHALL BE SUFFICIENT TO FILL THE VOID BETWEEN THE UNITIZED LOADS AND THE TRAILER SIDE WALL. PIECE 15 OF THE ASSEMBLY IS POSITIONED AGAINST THE TRAILER WALL. A SLIP FIT IS SATISFACTORY. TOENAIL TO TRAILER FLOOR WITH TWO 16d NAILS PER JOINT, TWO PLACES, TO PREVENT LONGITUDINAL MOVEMENT.
7. HALF LENGTH SIDE FRAME ASSEMBLIES, DETAILS E AND F, SHALL BE USED WHENEVER THE SIDE FRAME ASSEMBLIES, DETAILS C AND D, ARE TOO LONG.
8. WHEN TRAILERS HAVE NON-NAILABLE FLOORS AND SLEEPER PIECE 3 CAN NOT BE NAILED TO TRAILER FLOOR, HALF HEIGHT SIDE FRAME ASSEMBLIES (DETAILS C AND D MADE TO FIT ONE HIGH UNIT LOADS) MAY BE USED.
9. THE REAR BLOCKING ASSEMBLY, DETAIL B, SHALL FILL THE VOID BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.
10. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS AND ATTACH "EXPLOSIVE A" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
12. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
13. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL & LTL 40, 42, 44 & 45 FT TRAILERS (VAN)

WARNING

THIS STANDARD PROVIDES FOR INCREASED LOADS WHICH PRODUCE HEAVIER GROSS TANDEM AXLE WEIGHTS AND GROSS VEHICLE WEIGHTS. THESE INCREASED WEIGHTS ARE NOT AUTHORIZED IN ALL 50 STATES. BEFORE LOADING MORE THAN 14 UNIT LOADS IN A TRAILER, DETERMINE THAT THE LIMITATIONS OF THE NOTES IN TABLES III, IV, V AND VI ARE MET.

1. TO DETERMINE THE CORRECT LOAD PATTERN TO USE, SELECT THE CORRECT TABLE FOR THE LENGTH OF TRAILER BEING LOADED. OPPOSITE THE NUMBER OF UNIT LOADS TO BE LOADED, FIND THE CORRECT LOAD PATTERN. MAKE CERTAIN, THAT THE QUANITY BEING SHIPPED IS COMPATIBLE WITH THE WEIGHT OF THE TRACTOR-TRAILER BEING LOADED AND THE WEIGHT LAWS OF THE STATES THROUGH WHICH THE SHIPMENT IS ROUTED.

NOTE

LIGHT WEIGHT TRACTOR-TRAILERS: THE APPROXIMATE TARE WEIGHT OF THE TRAILER TANDEM AXLES IS 7,200 LBS AND OF THE DRIVE TANDEM AXLES 10,500 LBS.

STANDARD WEIGHT TRACTOR-TRAILERS: THE APPROXIMATE TARE WEIGHT OF THE TRAILER TANDEM AXLES IS 7,800 LBS AND OF THE DRIVE TANDEM AXLES 11,500 LBS.

THE ABOVE WEIGHTS MAY BE ADJUSTED UPWARD WHEN EXPERIENCE INDICATES THAT THE WEIGHT RESTRICTIONS CAN BE MET.

2. BLOCK AND BRACE THE SELECTED LOAD PATTERN USING THE PRINCIPLES OF THE TRUCKLOADS SHOWN ON PAGES 5 AND 6.
3. THE TRUCKLOAD SHOWN ON PAGES 5 AND 6 DO NOT LIMIT SHIPMENTS TO THE NUMBER OF UNIT LOADS SHOWN. THEY ARE PRESENTED TO SHOW THE BASIC PRINCIPLES OF LOADING.
4. WHEN SHIPPING LESS THAN THE AMOUNTS SHOWN IN TABLES II, IV, V AND VI, LOAD IN A SINGLE ROW PATTERN AND BLOCK AND BRACE FOLLOWING THE PRINCIPLES SHOWN ON PAGE 6. DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS.

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TABLE I

MK MOD AND TYPE	UNIT LOAD DWG	DIMENSIONS	CUBE (CU FT)	WEIGHT (LBS)	DOT HAZARD CLASSIFICATION
MK 83 MODS 0, 1, 2, 3, & 4 WITH PLASTIC NOSE PLUGS	WR-54/35	L - 74 1/2 W - 43 1/2 H - 19	35.5	2,935	EXPLOSIVES A
MK 83 MOD 5 THERMALLY PROTECTED	WR-54/243	L - 74 1/4 W - 43 1/2 H - 19 1/2	36.4	2,976	EXPLOSIVES A

TABLE II

TRUCKLOAD DATA

NO. OF UNIT LOADS	TRUCKLOAD WEIGHT (LBS) APPROX	
	MK 83 MODS 0, 1, 2, 3, 4	MK 83 MOD 5
16	46,960	47,616
15	44,025	44,640
14	41,090	41,664
13	38,155	38,688
12	35,220	35,712
11	32,285	32,736

TABLE III

40 FT TRAILER

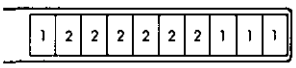
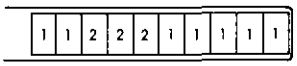
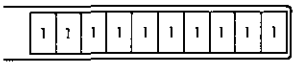
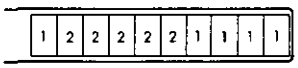
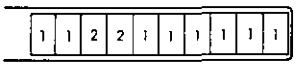
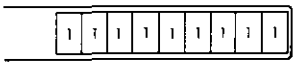
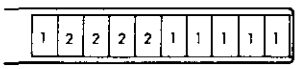
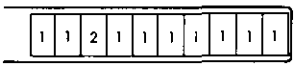
NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD
16**		13		10	
15*		12		9	
14		11			

TABLE IV

42 FT TRAILER

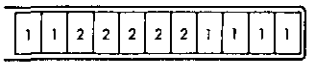
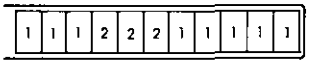
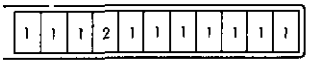
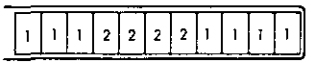
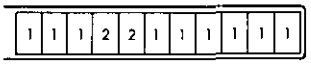
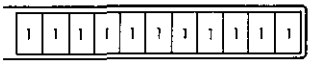
NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD
16**		14		12	
15*		13		11	

TABLE V

44 FT TRAILER

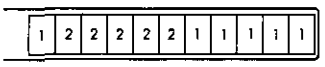
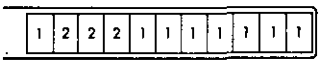
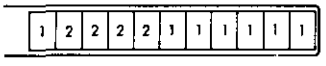
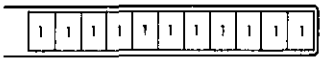
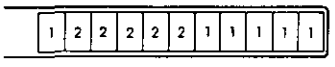
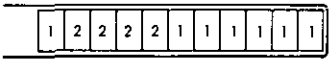
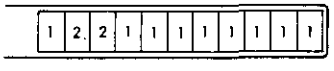
NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD
16**		14		12	
15*		13		11	

TABLE VI

45 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD	NO. OF UNIT LOADS	LOAD PATTERN → FWD
16**		14		12	
15*		13		11	

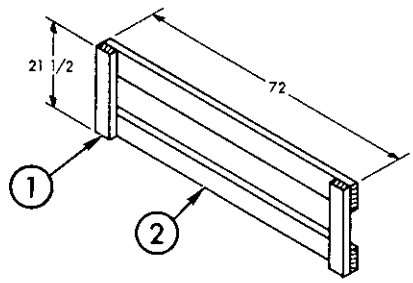
** AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS. ONLY LIGHT WEIGHT TRACTOR-TRAILERS SHALL BE USED.

* AUTHORIZED FOR LIGHT WEIGHT TRACTOR-TRAILERS ONLY WHEN ROUTING IS THROUGH STATES LIMITING TANDEM AXLES TO 32,000 LBS AND GROSS VEHICLE WEIGHTS OF 73,280 LBS. AUTHORIZED FOR BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS ON TANDEM AXLES AND GROSS VEHICLE WEIGHTS OF 80,000 LBS.

NOTE: WHEN SHIPPING 14 OR LESS UNIT LOADS, BOTH STANDARD AND LIGHTWEIGHT TRACTOR-TRAILERS MAY BE USED IN ALL 50 STATES WITHOUT EXCEEDING THE WEIGHT LAWS.

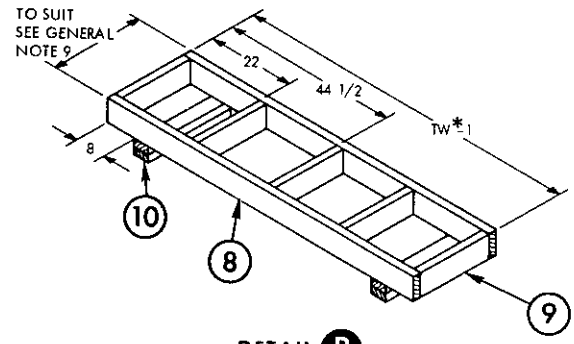
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* TW = TRAILER WIDTH



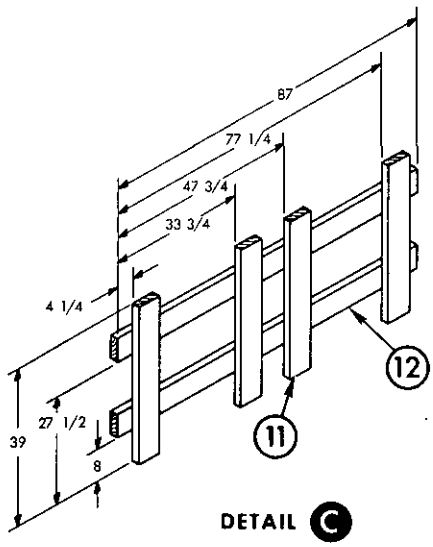
DETAIL A

FRONT BULKHEAD
(1 REQUIRED)



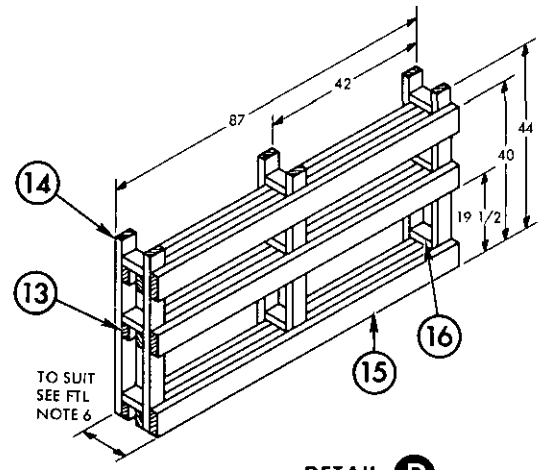
DETAIL B

REAR BLOCKING ASSEMBLY



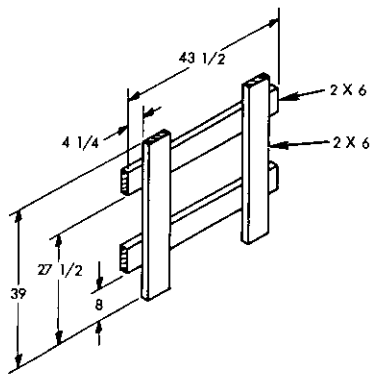
DETAIL C

SIDE FRAME ASSEMBLY
(NOSE END)



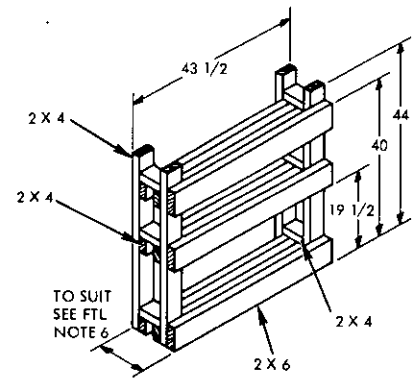
DETAIL D

SIDE FRAME ASSEMBLY
(BASE END)



DETAIL E

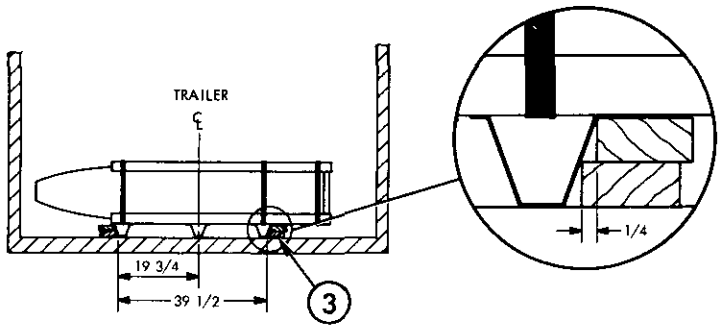
HALF LENGTH SIDE FRAME ASSEMBLY
(NOSE END)
SEE GENERAL NOTE 7



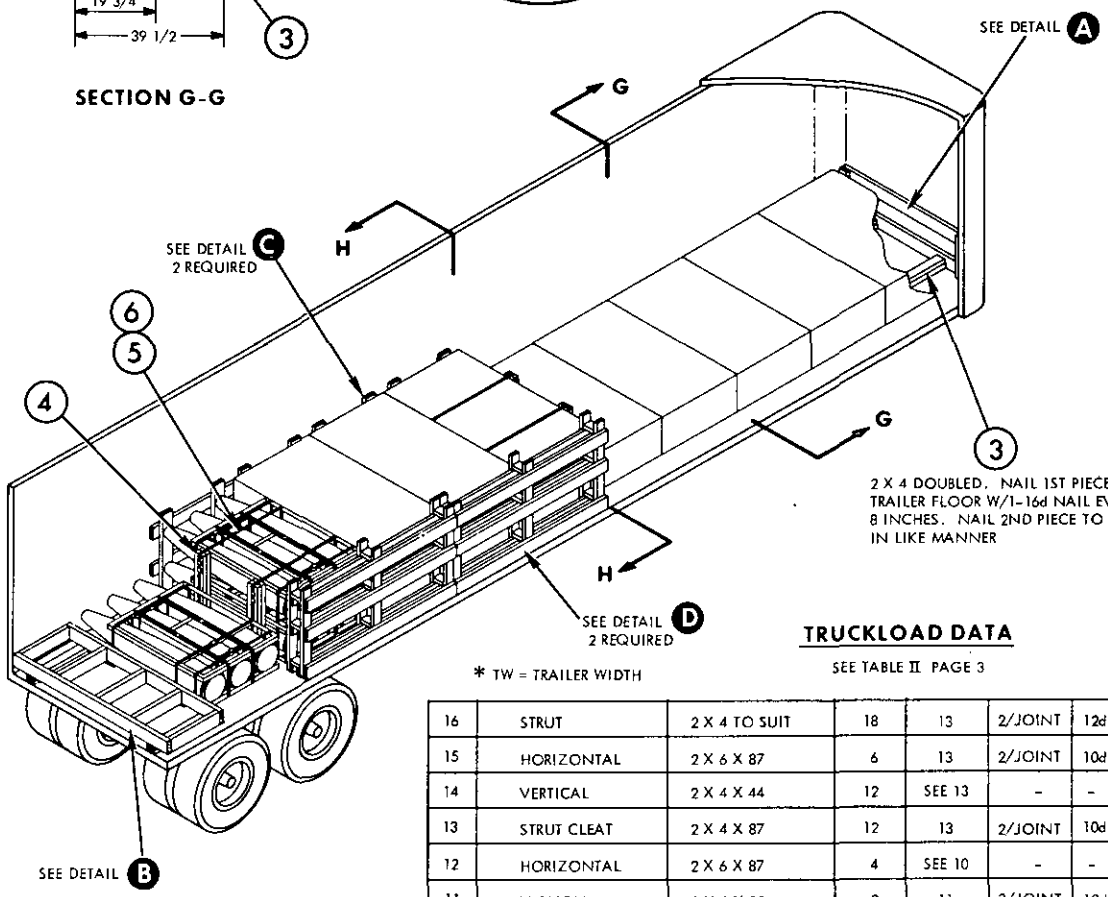
DETAIL F

HALF LENGTH SIDE FRAME ASSEMBLY
(BASE END)
SEE GENERAL NOTE 7

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SECTION G-G



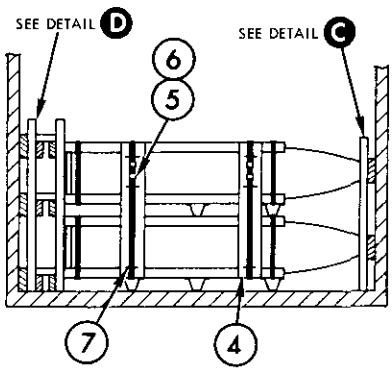
2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER

TRUCKLOAD DATA

SEE TABLE II PAGE 3

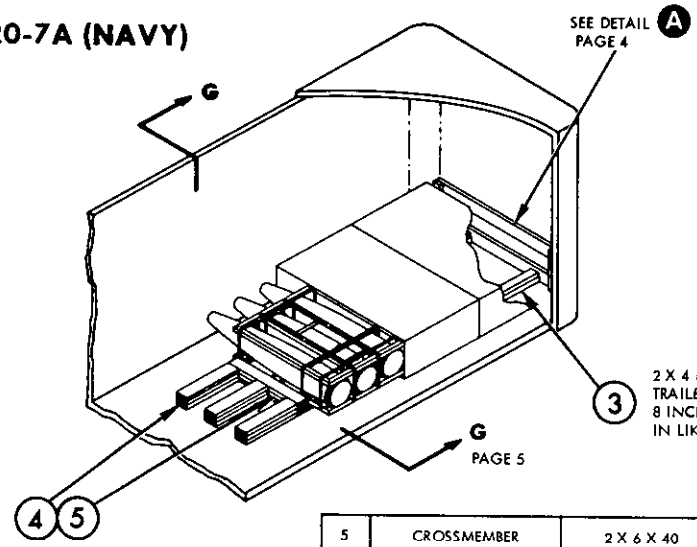
* TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS: NUMBER	NAILS: SIZE
	LIST OF MATERIALS & NAILING DATA					
16	STRUT	2 X 4 TO SUIT	18	13	2/JOINT	12d
15	HORIZONTAL	2 X 6 X 87	6	13	2/JOINT	10d
14	VERTICAL	2 X 4 X 44	12	SEE 13	-	-
13	STRUT CLEAT	2 X 4 X 87	12	13	2/JOINT	10d
12	HORIZONTAL	2 X 6 X 87	4	SEE 10	-	-
11	VERTICAL	2 X 6 X 39	8	11	3/JOINT	10d
10	SUPPORT	2 X 4 TO SUIT	4	8 10	2/JOINT 3/PIECE	10d 10d
9	STRUT	2 X 6 TO SUIT	5	SEE 8	-	-
8	CROSSMEMBER	2 X 6 X TW* 1	2	9	3/JOINT	16d
7	STAPLE	FOR 1 1/4 STRAP	12	-	-	-
6	SEAL	FOR 1 1/4 STRAP	8	-	-	-
5	STRAP	1 1/4 X .035 X 15 FT	4	-	-	-
4	STIFFENER	2 X 6 X 35	8	DOUBLED LAMINATE	4	10d
3	SLEEPER	2 X 4 X 43	24	SEE FIELD NOTE		
2	HORIZONTAL	2 X 6 X 72	2	SEE 1	-	-
1	VERTICAL	2 X 4 X 21 1/2	2	2	3/JOINT	10d



SECTION H-H

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2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

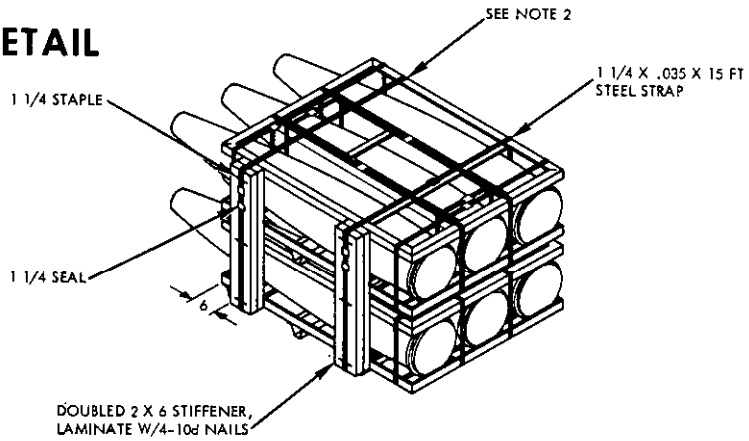
2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
5	CROSSMEMBER	2 X 6 X 40	3	SEE FIELD NOTE		
4	BACKUP CLEAT	2 X 6 X 30	9	SEE FIELD NOTE		
3	SLEEPER	2 X 4 X 43	12	SEE FIELD NOTE		
2	HORIZONTAL	2 X 6 X 72	2	SEE 1	-	-
1	VERTICAL	2 X 4 X 21 1/2	2	2	3/JOINT	10d

LIST OF MATERIALS & NAILING DATA

LTL ALL TRAILERS

UNITIZING DETAIL



1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS MILITARY STANDARD, THE TWO-HIGH STACKED UNIT LOADS SHALL BE UNITIZED AS SHOWN ABOVE.
2. THE DOUBLED 2 X 6 STIFFENERS SHALL EXTEND FROM THE TOP OF THE STACKED UNIT LOADS TO THE BOTTOM EDGE OF THE LOWER FRAME. WHEN THERE IS ONLY ONE UNIT LOAD IN THE SECOND LAYER, STIFFENERS ARE REQUIRED ON BOTH SIDES OF THE UNITIZED LOADS.
3. THE 1 1/4 STEEL STRAPS, POSITIONED AS SHOWN, ENCIRCLED THE STACKED UNIT LOADS AND PASS UNDER THE BOTTOM OF THE LOWER FRAME. THE STRAPS HOLD THE STIFFENERS IN PLACE.
4. TENSION STRAPS AND SEAL WITH TWO DOUBLED CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH THREE 1 1/4 INCH STAPLES.

REVIEW ACTIVITIES:
NAVY - OS, AS

PREPARING ACTIVITY:
NAVY - OS
PROJECT NO. 8140-N161

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